

TP3: Browser Fingerprinting

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Task 0 – HTTP Request Headers

Header	Information revealed
User-Agent	Reported browser, version, and operating system.
Accept-Language	Preferred languages and their priorities.
Accept	Accepted response formats.
Accept-Encoding	Supported compression formats.
Sec-CH-UA	Browser brands and major versions.
Sec-CH-UA-Platform	Reported operating-system platform.
Sec-CH-UA-Mobile	Whether the browser reports a mobile device.
DNT	Do Not Track preference; does not enforce blocking.
Sec-Fetch-Site	Origin relationship between requester and destination.
Sec-Fetch-Mode	Request mode, such as navigation.

Combining browser, language, platform, and capability values can help distinguish configurations; fetch metadata describes the current request.

Task 1 – Passive Collector

The Flask route was extended with the eight additional headers listed after **Accept-Language** above. Normal Edge and Edge InPrivate returned the same values: Edge/Chromium 154, Windows, mobile flag ?0, DNT: 1, **Sec-Fetch-Site: none**, and **Sec-Fetch-Mode: navigate**. The language preferences, accepted formats, and compression list (gzip, deflate, br, zstd) were also unchanged.

```
Windows PowerShell
PS C:\Users\User\Desktop> .\Info Sec\TP3\FingerprintingLab\app.py, reloading
* Detected change in 'C:\Users\User\Desktop\Info Sec\TP3\FingerprintingLab\app.py', reloading
* Restarting with stat
* Debugger is active!
* Debugger PIN: 119-528-873
* Detected change in 'C:\Users\User\Desktop\Info Sec\TP3\FingerprintingLab\app.py', reloading
* Restarting with stat
* Debugger is active!
* Debugger PIN: 119-528-873

===== NEW VISIT =====
IP address      127.0.0.1
HTTP method     GET
User-Agent      Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0 Safari/537.36 Edg/154.0.0
Accept-Language fr-FR;q=0.9,en;q=0.8,de;q=0.7,en-US;q=0.6
Accept          text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding gzip, deflate, br, zstd
Sec-CH-UA       "Chromium";v="154", "Microsoft Edge";v="154", "Not A(Brand)";v="99"
Sec-CH-UA-Platform "Windows"
Sec-CH-UA-Mobile 1
DNT             1
Sec-Fetch-Site  none
Sec-Fetch-Mode  navigate

127.0.0.1 - - [29/Sep/2026 11:55:03] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [29/Sep/2026 11:55:03] "GET / wsl://wsl/~/Documents/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -

===== NEW VISIT =====
IP address      127.0.0.1
HTTP method     GET
User-Agent      Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0 Safari/537.36 Edg/154.0.0
Accept-Language fr-FR;q=0.9,en;q=0.8,de;q=0.7,en-US;q=0.6
Accept          text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding gzip, deflate, br, zstd
Sec-CH-UA       "Chromium";v="154", "Microsoft Edge";v="154", "Not A(Brand)";v="99"
Sec-CH-UA-Platform "Windows"
Sec-CH-UA-Mobile 1
DNT             1
Sec-Fetch-Site  none
Sec-Fetch-Mode  navigate

127.0.0.1 - - [29/Sep/2026 11:55:22] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [29/Sep/2026 11:55:22] "GET / wsl://wsl/~/Documents/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -
```

Figure 1: Passive headers received and printed by Flask.

Conclusion: InPrivate did not hide the selected passive HTTP features in this experiment.

Task 2 – Active Feature Collection

JavaScript collected five feature groups, displayed them, and sent them as JSON to `/collect`. Flask printed the received values.

Feature	Information and usefulness
navigator.language navigator. hardwareConcurrency screen.width/height window.innerWidth/ innerHeight Intl.DateTimeFormat(). resolvedOptions().timeZone	Preferred language; distinguishes browser configurations. Reported logical processor count; hardware-related clue. Screen dimensions in CSS pixels; display-related clue. Viewport dimensions; describes window layout. Configured time zone; regional configuration clue.

Feature	Edge	Brave
Language	fr	fr-FR
Logical processors	4	4
Screen dimensions	1280 × 720	1280 × 720
Viewport dimensions	321 × 624	278 × 565
Time zone	Africa/Casablanca	Africa/Casablanca

```

Windows PowerShell
127.0.0.1 ~ [29/Sep/2026 12:21:52] "GET /static/fingerprint.js HTTP/1.1" 200 -
===== ACTIVE FEATURES =====
Language      : fr-FR
CPU logical cores : 4
Screen resolution : 1280x720
Window size    : 1280x565
Time zone      : Africa/Casablanca
=====

127.0.0.1 ~ [29/Sep/2026 12:21:53] "POST /collect HTTP/1.1" 200 -
===== NEW VISIT =====
IP address      : 127.0.0.1
HTTP method     : GET
User-Agent       : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36 Edg/154.0.0.0
Accept-Language : fr-FR;q=0.9,en;q=0.8,en-GB;q=0.7,en-US;q=0.6
Accept          : text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8,application/signed-exchange;v=b3;q=0.7
Accept-Encoding : gzip, deflate, br, rzd
Sec-CH-UA       : "Chromium";v="154", "Microsoft Edge";v="154", "Not A(Brand";v="99"
Sec-CH-UA-Platform : "Windows"
Sec-CH-UA-Mobile : 0
DNT             : 1
Sec-Fetch-Site  : none
Sec-Fetch-Mode  : navigate
=====

127.0.0.1 ~ [29/Sep/2026 12:21:56] "GET / HTTP/1.1" 200 -
127.0.0.1 ~ [29/Sep/2026 12:21:59] "GET /well-known/appspecific/csm.chrome.devtools.json HTTP/1.1" 404 -
127.0.0.1 ~ [29/Sep/2026 12:23:41] "GET /well-known/appspecific/csm.chrome.devtools.json HTTP/1.1" 404 -
===== NEW VISIT =====
IP address      : 127.0.0.1
HTTP method     : GET
User-Agent       : Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/154.0.0.0 Safari/537.36
Accept-Language : fr-FR;q=0.9,en-GB;q=0.8,en;q=0.7
Accept          : text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8
Accept-Encoding : gzip, deflate, br, rzd
Sec-CH-UA       : "Chromium";v="154", "Brave";v="154", "Not A(Brand";v="99"
Sec-CH-UA-Platform : "Windows"
Sec-CH-UA-Mobile : 0
DNT             : 1
Sec-Fetch-Site  : none
Sec-Fetch-Mode  : navigate
=====

127.0.0.1 ~ [29/Sep/2026 12:23:45] "GET / HTTP/1.1" 200 -
127.0.0.1 ~ [29/Sep/2026 12:23:45] "GET /well-known/appspecific/csm.chrome.devtools.json HTTP/1.1" 404 -
127.0.0.1 ~ [29/Sep/2026 12:23:46] "GET /static/fingerprint.js HTTP/1.1" 304 -
===== ACTIVE FEATURES =====
Language      : fr-FR
CPU logical cores : 4
Screen resolution : 1280x720
Window size    : 278x565
Time zone      : Africa/Casablanca
=====

127.0.0.1 ~ [29/Sep/2026 12:23:46] "POST /collect HTTP/1.1" 200 -

```

Figure 2: Active features received by Flask through POST `/collect`.

The processor count, screen dimensions, and time zone matched. Language differed between browsers. Viewport size depends on the window and Developer Tools layout; the terminal also shows an earlier, wider viewport sample.

Task 2 – Protection Comparison

In Edge with client-side blocking, `fingerprint.js` failed with `ERR_BLOCKED_BY_CLIENT`; no active values were collected. Renaming the same script to `features.js` allowed it to load and send the values. This suggests a name/URL-based blocking rule in the tested configuration.

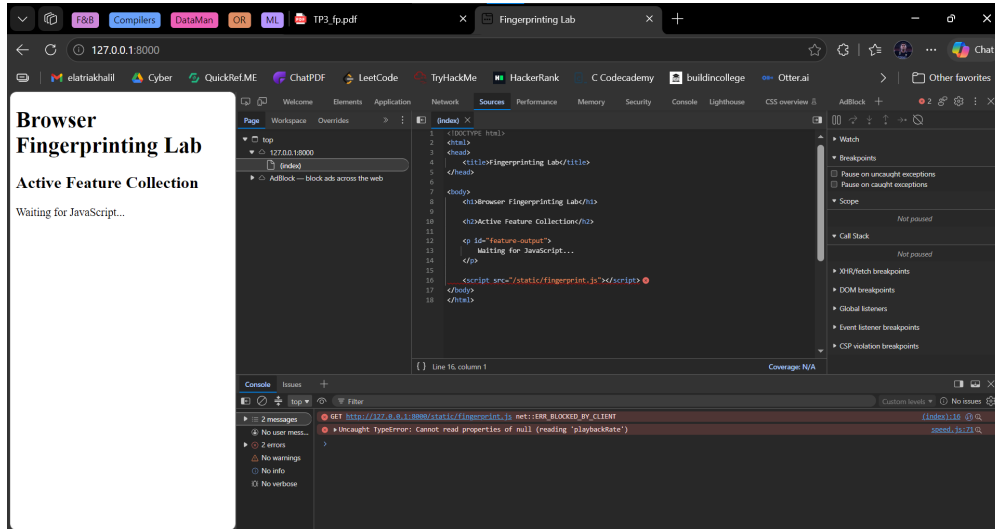


Figure 3: Original fingerprint.js blocked before execution.

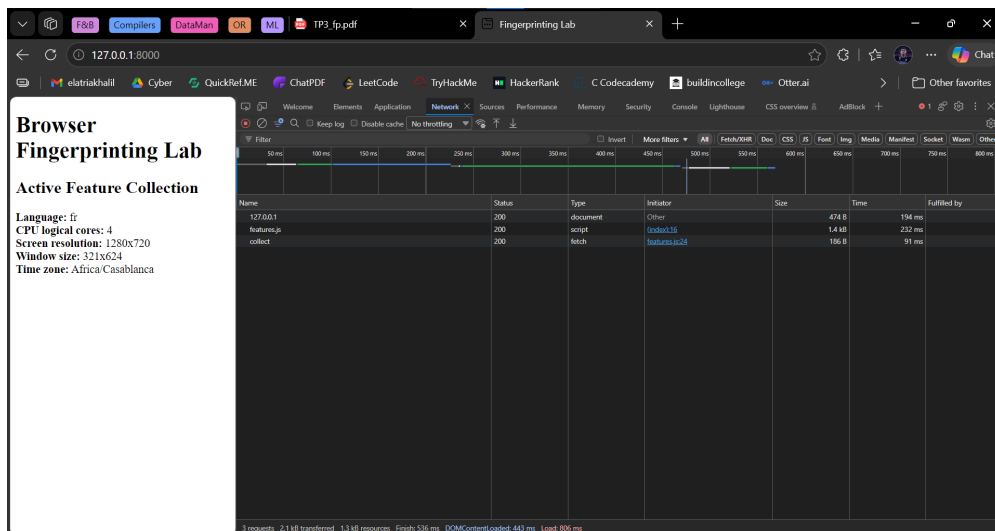


Figure 4: Renamed features.js and /collect both returning HTTP 200.

Edge InPrivate still exposed the active values. Private mode did not prevent the collector from executing in this test. **Conclusion:** Script blocking was the most effective observed protection against this active collector, but renaming bypassed the block. Passive headers remained available. Cookie blocking was not separately tested, and the local results do not establish an overall privacy ranking of Edge and Brave.

Task 3 – Typing Interface

The interface contains the target sentence, text input, Restart button, and empty result area, with IDs `target-sentence`, `typing-input`, `restart-button`, and `typing-results`.

Cookies are gone, but Alice's rhythm still leaves traces!

Task 4 – Alice's Typing Rhythm

Formula: typing speed is the target length divided by elapsed time:

$$T = \frac{t_{\text{end}} - t_{\text{start}}}{1000}, \quad v = \frac{57}{T} \text{ characters/second.}$$

Algorithm: `keydown` starts the timer on the first character and counts Backspace presses. `input` checks for an exact match, stops timing, calculates the speed, displays the results, and sends them to `/typing`. Restart clears the input, results, timer, and counter.

The recorded attempt took **74.12 seconds**, with **13 corrections**. Its speed was $57/74.12 \approx 0.77$ characters/second.

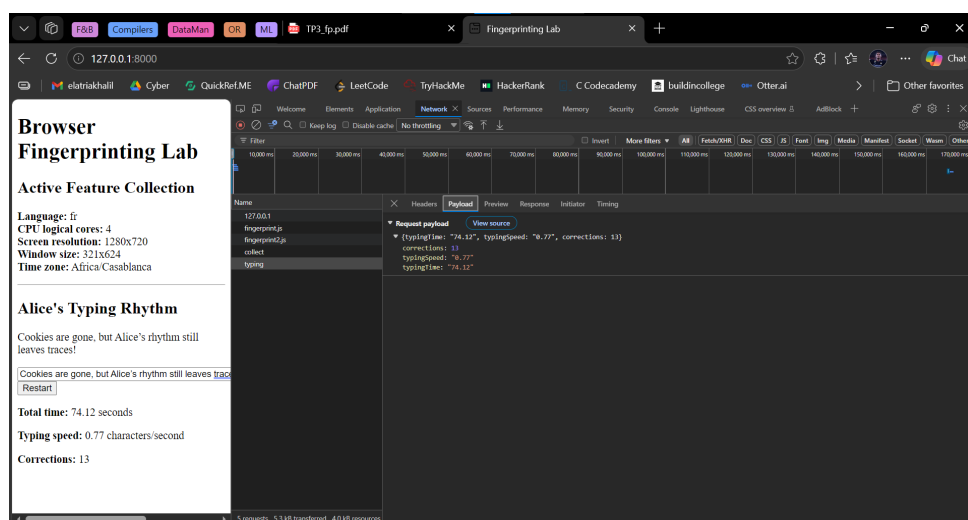


Figure 5: Displayed results and the `/typing` JSON payload.

```
127.0.0.1 -- [29/Sep/2026 22:12:15] "GET / HTTP/1.1" 200 -
127.0.0.1 -- [29/Sep/2026 22:12:15] "GET /.well-known/appspecific/com.chrome.devtools.json HTTP/1.1" 404 -
127.0.0.1 -- [29/Sep/2026 22:12:15] "GET /static/fingerprint2.js HTTP/1.1" 200 -
127.0.0.1 -- [29/Sep/2026 22:12:15] "GET /static/fingerprint.js HTTP/1.1" 200 -

===== ACTIVE FEATURES =====
Language       : fr
CPU logical cores : 4
Screen resolution : 1280x720
Window size    : 321x624
Time zone      : Africa/Casablanca
=====

127.0.0.1 -- [29/Sep/2026 22:12:15] "POST /collect HTTP/1.1" 200 -

===== TYPING BEHAVIOUR =====
Typing time    : 74.12
Typing speed   : 0.77
Corrections    : 13
=====

127.0.0.1 -- [29/Sep/2026 22:15:00] "POST /typing HTTP/1.1" 200 -
```

Figure 6: Flask receiving the same measurements and returning HTTP 200.

Conclusion: The behavioural measurements were successfully displayed and transmitted. One sample does not establish identity: typing time and corrections can vary with pauses, fatigue, keyboard, and environment.